

Machine Learning

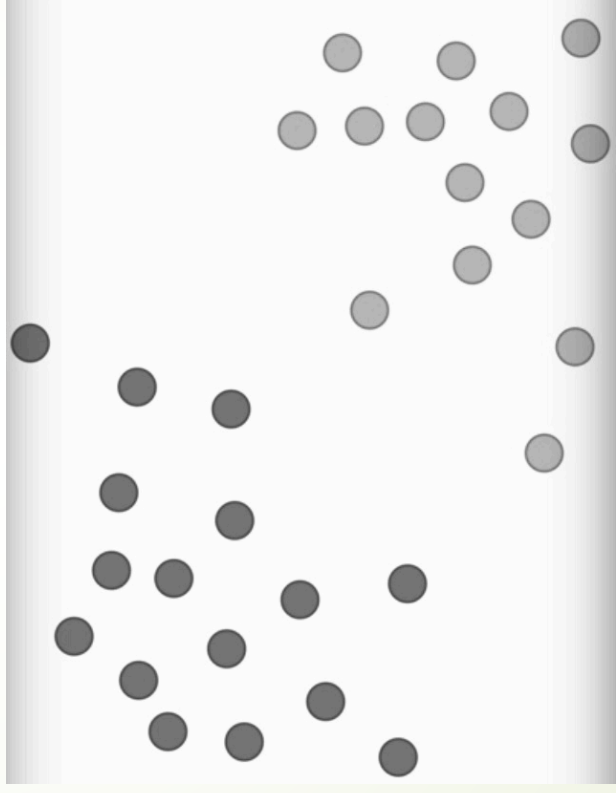
Lecture 6

Supervised Learning Techniques and Algorithms: Support Vector
Machines

The majority of this content is taken from
Alice Zhao lectures:
<https://github.com/adashofdata>

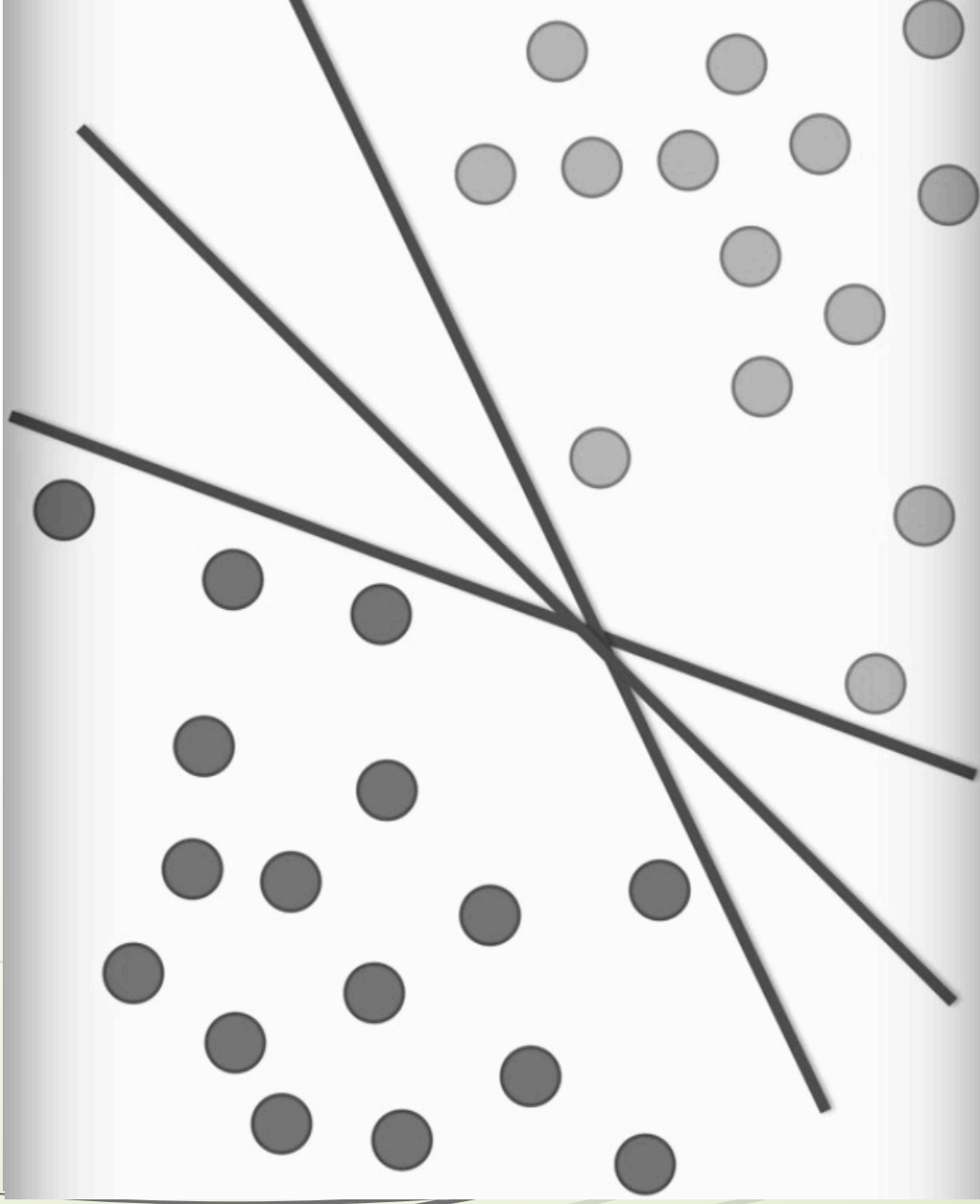
What is SVM

- What is SVM?
- It is a classification method.
- What does it mean?
- I have two sets of data (black dots and cyan dots) and I want to classify them (split them into two sets)



What is SVM

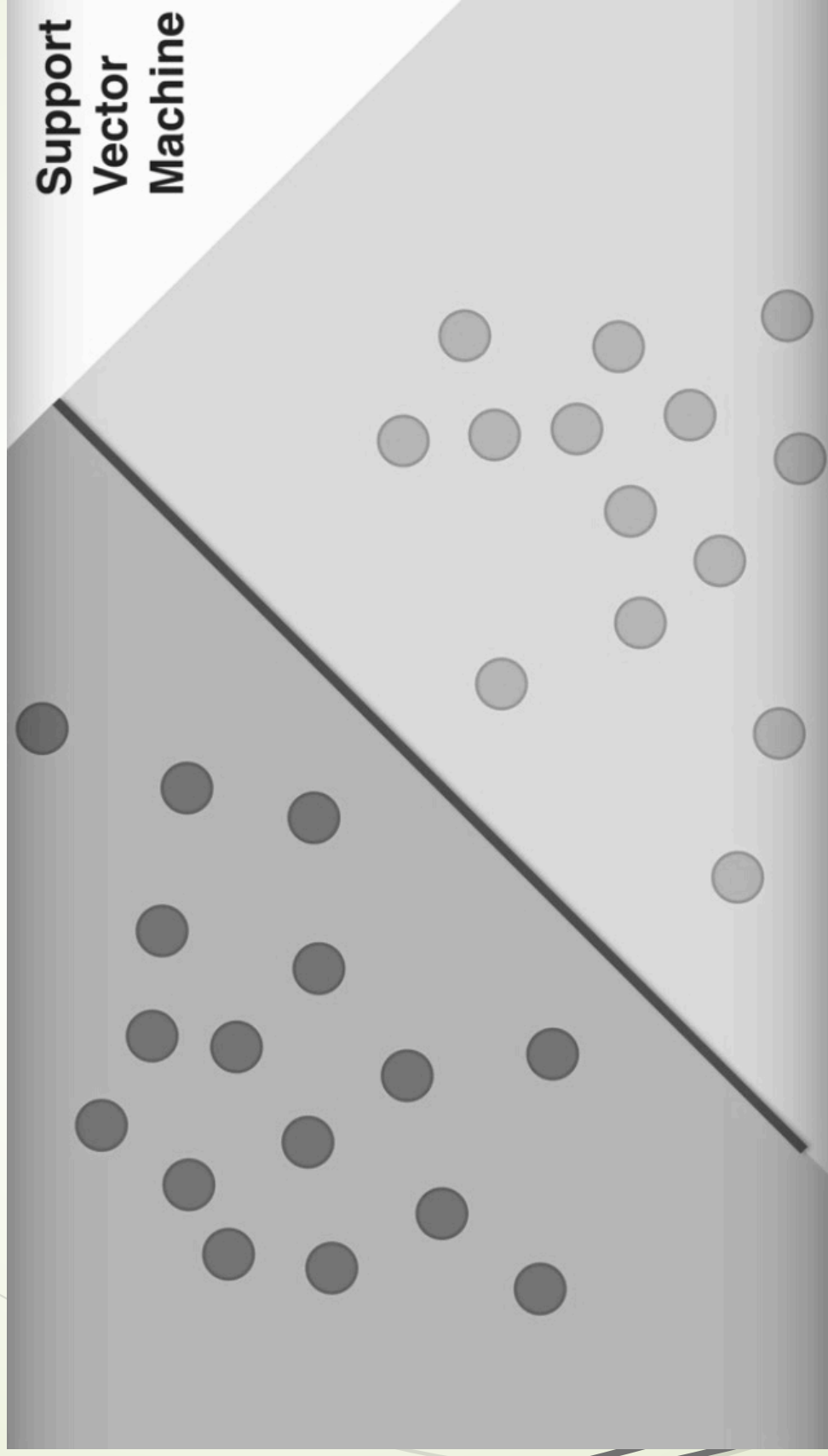
- Which one is the best way to split the data?



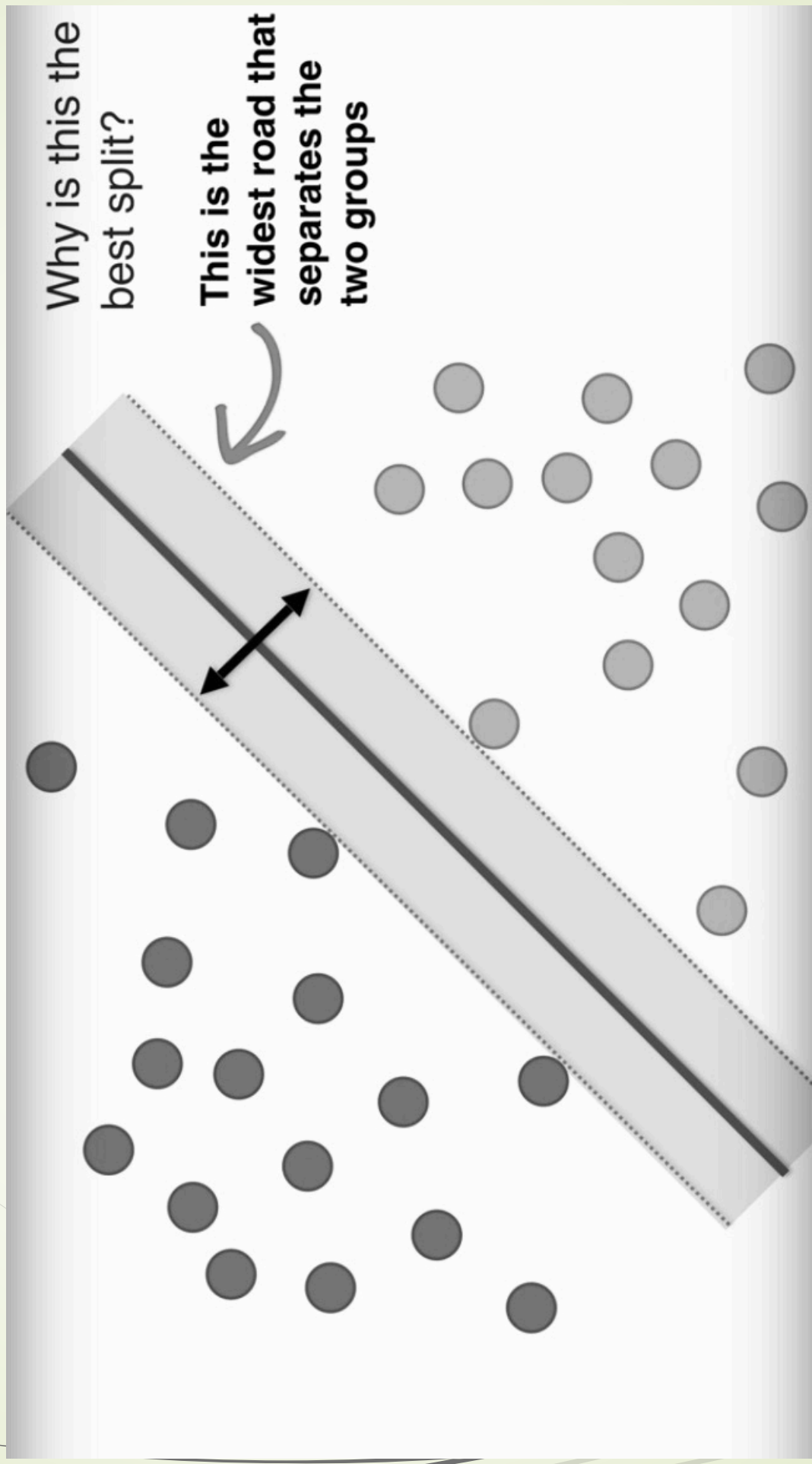
All of these
splitting are fine
but what if we
seek the best
split?

What is SVM

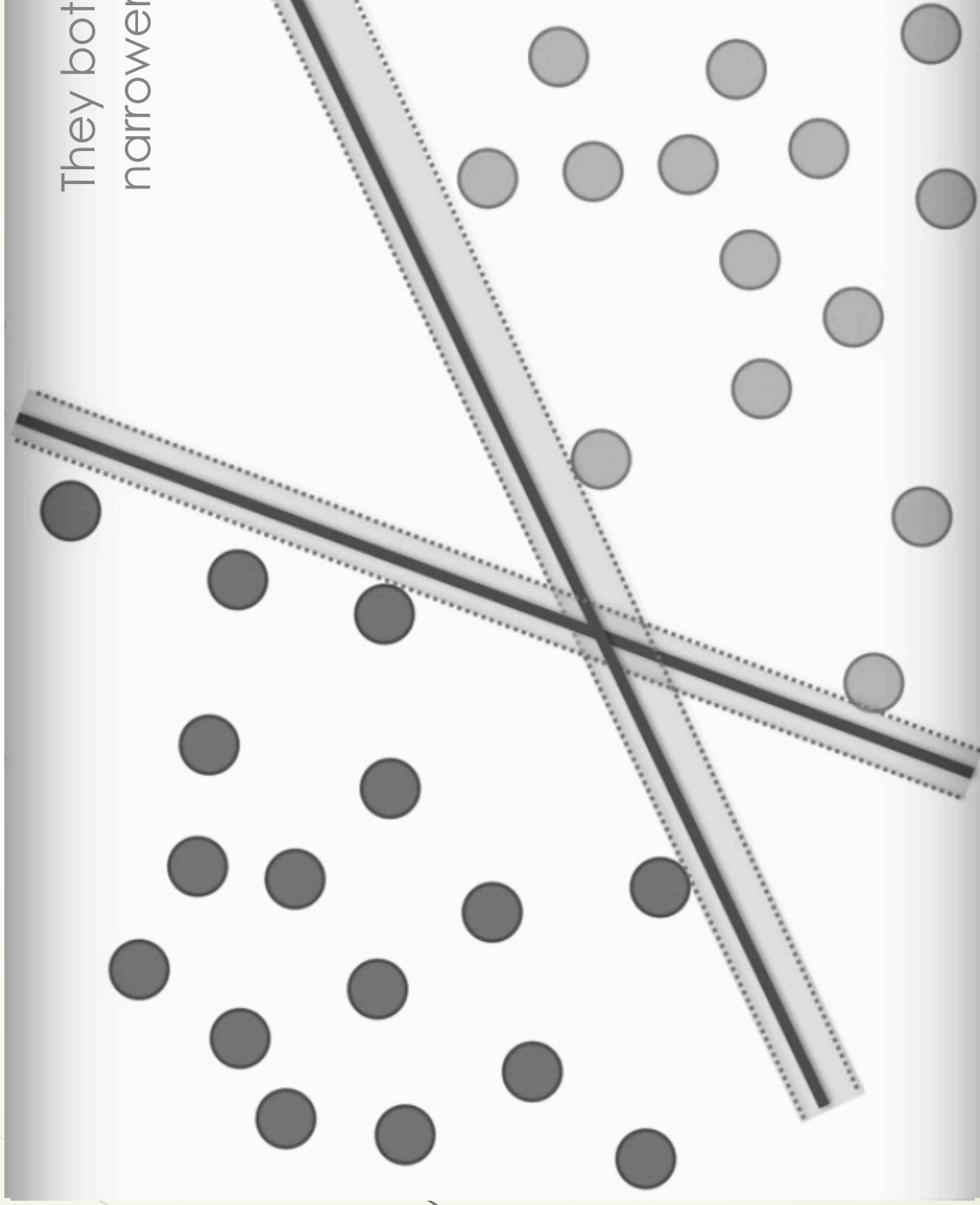
- ▶ I claim this is the best line (SVM)

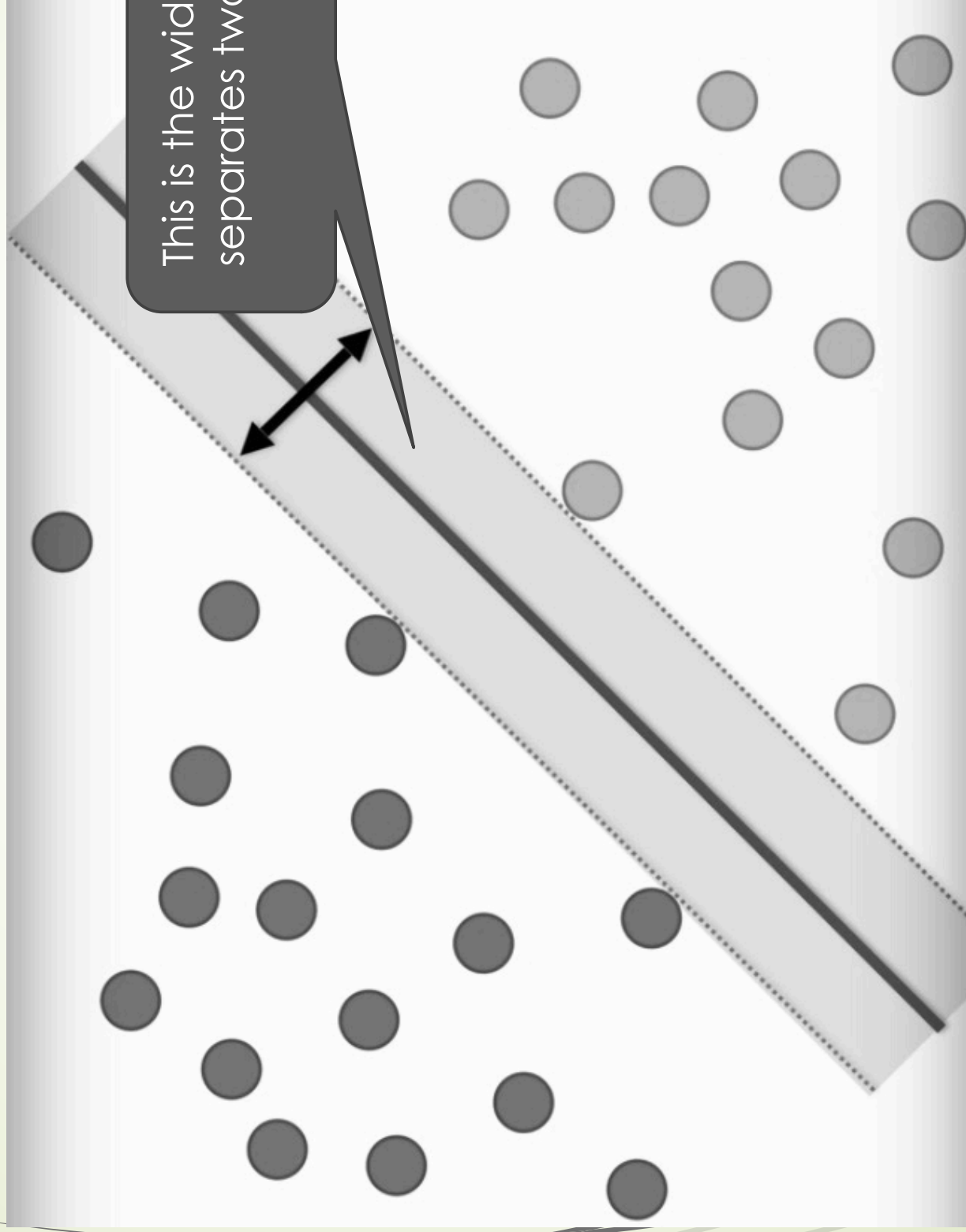


Why is this the best?



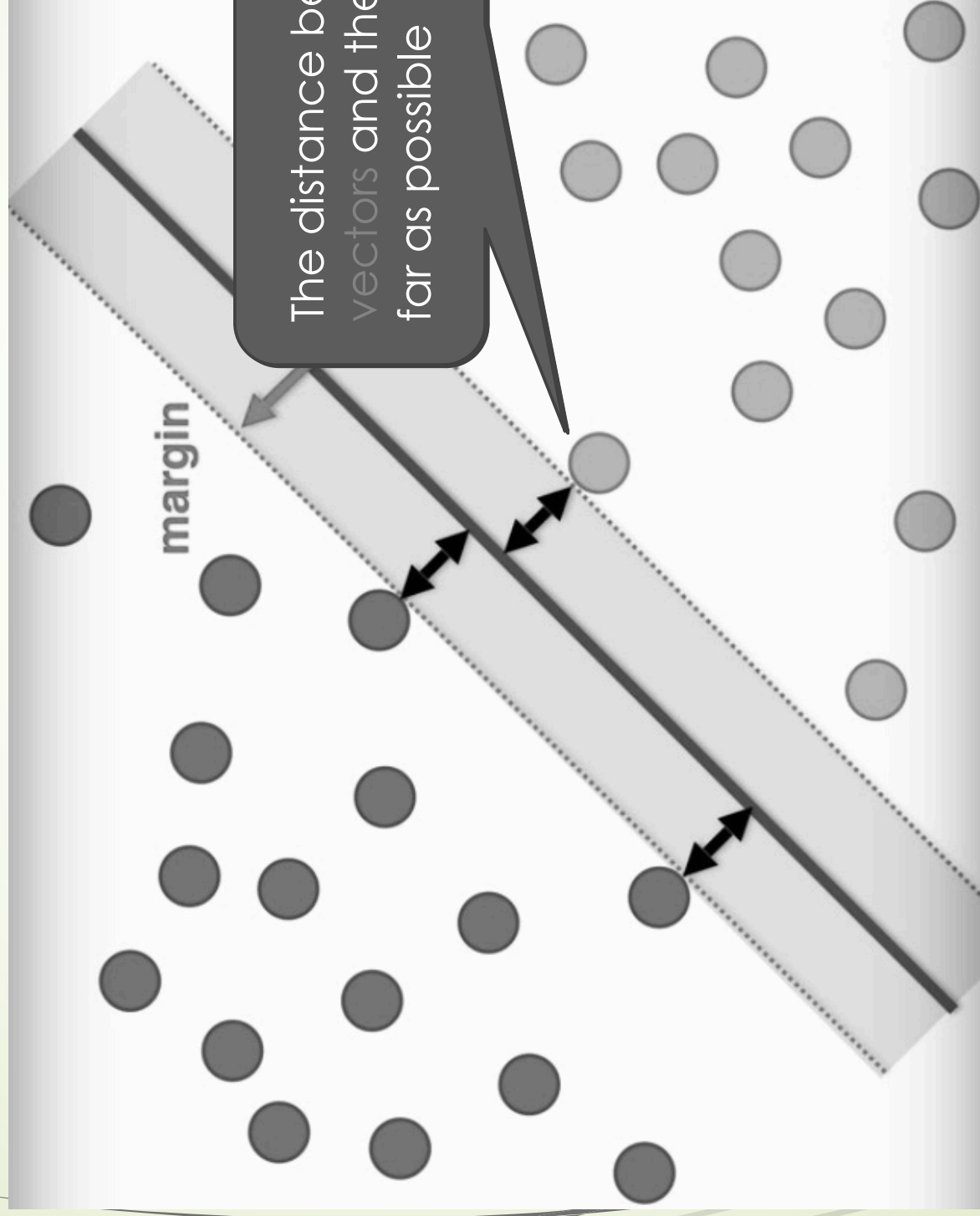
How about other ones?





Technical terms

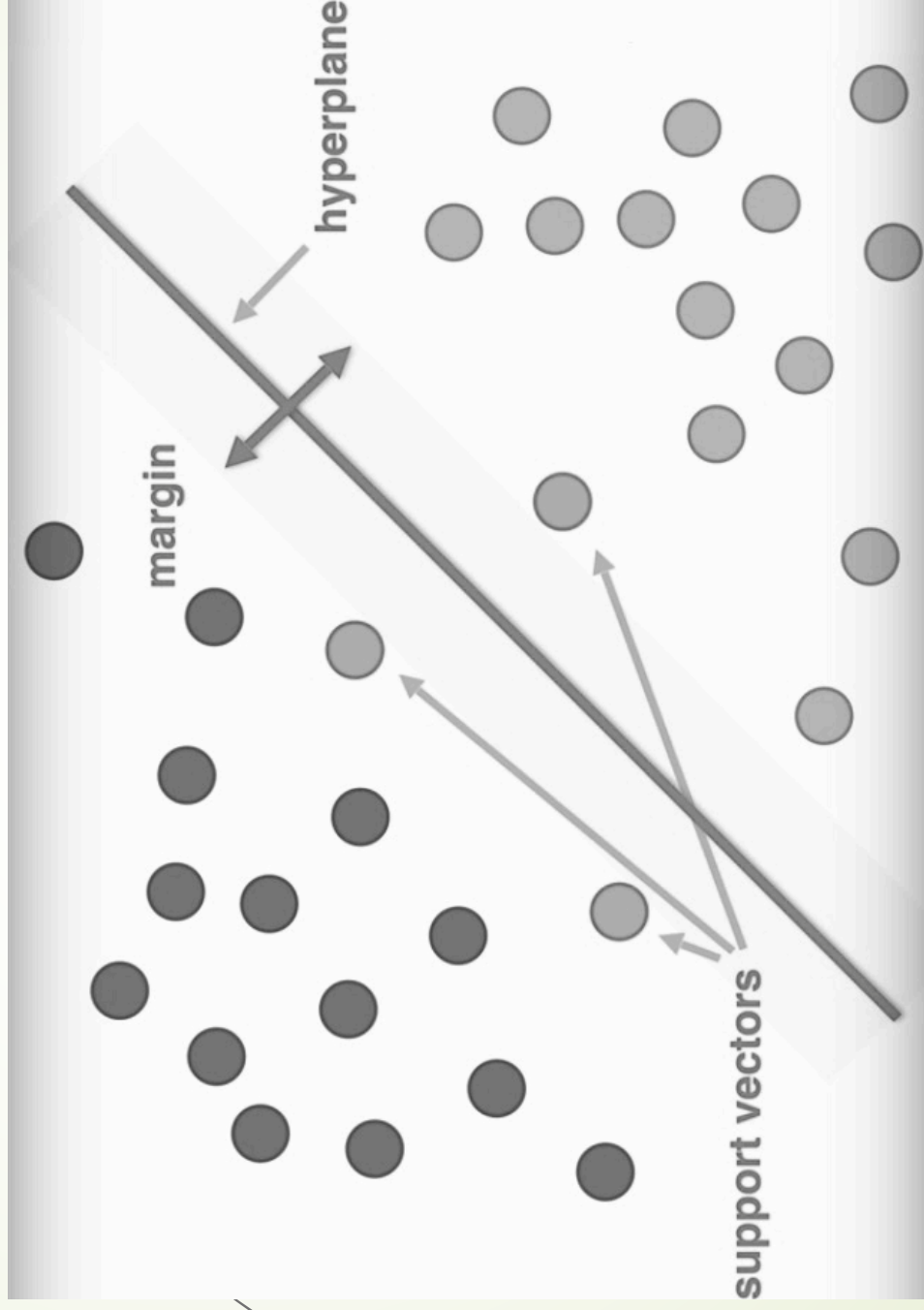
8



In conclusion

9

- I found a hyperplane that best splits the data
- It is as far as possible from vector machines
- Or I can say I maximized the margins



Linear Optimization

10

- Let's have a review on optimization
- Identify the decision variables
- Identify the objective function
- Identify all appropriate constraints
- Write the objective function and constraints as mathematical expressions